

Based on the RFP, the project is to build an **Enterprise Data Lakehouse, Governance and Analytics Platform** for OICL. The RFP expects the bidder (System Integrator/OEM) to propose an **indicative architecture** that covers ingestion, storage, governance, analytics, AI, security, APIs, and consumption layers. The project objective is to create a centralized platform for fraud detection, business intelligence, AI/ML, and enterprise data sharing.

Indicative Architecture

OICL USERS

Business Users | Fraud Team | Data Scientists | Executives | External Systems

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Visualization & Analytics Layer

- Dashboards (BI)
- Reports
- Self-Service Analytics
- Fraud Detection
- AI/ML Predictions
- Data APIs

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AI / Analytics / Processing Layer

- SQL Analytics
- Batch Processing
- Stream Processing
- ML Workbench

- Feature Store
- Model Training
- Explainable AI



Enterprise Data Lakehouse

Bronze Zone (Raw Data)

Silver Zone (Cleaned Data)

Gold Zone (Business Ready Data)

- Structured Data
- Semi-Structured Data
- Unstructured Documents
- Historical Archive



Governance & Metadata Layer

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- Data Catalog
 - Metadata Repository
 - Data Lineage
 - Data Quality
 - Master Data Management (MDM)
 - Data Classification
 - Audit Logs



Data Ingestion & Integration Layer

- Batch ETL/ELT
 - Streaming (Kafka/PubSub)
 - CDC
 - APIs
 - File Upload
 - Data Validation
 - Data Contracts
 - Replay / Backfill
-



OICL Source Systems

Core Insurance (INLIAS)

HRMS

SAP

CRM

Email

ECM

Claims

Policy Systems

Payment Gateway

External APIs

Government Databases

TPA

Hospitals

Police

IIB

Other Third Parties

Cross-Cutting Security Services

IAM • RBAC • MFA • Encryption • KMS/HSM

Data Masking • SIEM • SOAR • DLP

Audit • Compliance • Monitoring

Infrastructure Layer

Private Cloud / Public Cloud Ready

Containers

Virtual Machines

Object Storage

Compute Cluster

Backup

Disaster Recovery

Load Balancer

Monitoring

Layer-wise explanation

1. Source Systems

These are all the systems from which data is collected.

Examples:

- Core Insurance (INLIAS)
- Claims
- Policy Administration
- HRMS
- SAP
- ECM
- Email
- Customer Portal
- External agencies
- Hospitals
- Police
- TPAs
- Government databases

The RFP also expects integration with external regulators and agencies for fraud detection.

2. Data Ingestion Layer

Purpose:

Bring data into the platform.

Supports

- Batch ingestion
- Real-time streaming
- APIs
- File uploads
- CDC (Change Data Capture)

- Data contracts
 - Metadata capture
 - Replayability
 - Idempotency
-

3. Data Lakehouse

Central storage.

Contains

Bronze

- Raw data

↓

Silver

- Cleansed data

↓

Gold

- Business-ready datasets

Stores

- Structured
- Semi-structured
- Unstructured documents

The RFP expects support for schema-on-read and schema-on-write. (See Sections 2.3.2 and 2.3.7.4 in the table of contents.)

4. Governance Layer

Responsible for

- Metadata
- Data Catalog
- Data Quality

- Data Lineage
- MDM
- Audit
- Compliance

This ensures trusted, governed, and traceable data across the enterprise. The project objective specifically calls out data governance, metadata management, data quality, audit trails, and Master Data Management.

5. Processing Layer

Processes the data.

Includes

- Spark
- SQL Engine
- Stream Processing
- AI Engine
- ML Workbench
- Feature Engineering
- Model Training

Supports

- Fraud Detection
 - Risk Scoring
 - Predictive Analytics
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6. Analytics Layer

Used by business users.

Includes

- Power BI / Tableau
- Reports
- Dashboards

- KPI Monitoring
 - Self-Service BI
 - Fraud dashboards
 - Executive dashboards
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7. API Layer

Exposes data securely.

Provides

- REST APIs
- Master Data APIs
- Analytics APIs
- Data Exchange Hub

The project objective also requires data APIs and a common data exchange hub for real-time consumption.

8. Security Layer

Applied across every layer.

Includes

- IAM
- RBAC
- MFA
- Encryption
- Data Masking
- SIEM
- SOAR
- DLP
- Audit Logging
- KMS/HSM

9. Infrastructure Layer

Runs everything.

Supports

- Private Cloud
- Public Cloud Ready
- Containers
- Virtual Machines
- Compute
- Storage
- Backup
- Disaster Recovery
- High Availability

The RFP explicitly requires the architecture to be cloud-ready/cloud-native and deployable on private and public cloud.

As a Technical/Techno-Commercial Engineer

When reviewing an OEM's proposed architecture, ensure it clearly demonstrates:

Layer	What to verify
Source Systems	All OICL applications and external agencies are connected
Ingestion	Batch, streaming, CDC, replay, metadata, data quality
Lakehouse	Bronze/Silver/Gold zones, schema-on-read & write
Governance	Catalog, lineage, MDM, audit, metadata
Processing	Batch, streaming, AI/ML, SQL analytics
Analytics	Dashboards, BI, fraud detection, reporting
API	Secure APIs and data exchange hub
Security	IAM, RBAC, MFA, encryption, masking, SIEM, DLP
Infrastructure	Cloud-ready, HA, DR, scalability, monitoring

This is the kind of **end-to-end architecture diagram** that OEMs typically include in their technical proposal and that evaluators expect to see aligned with the RFP's architecture, governance, security, and analytics requirements.

For this RFP, the architecture should follow a **layered model**, where only the **Consumption Layer** is exposed to users, while all data storage and processing remain within the **Government Community Cloud (GCC)**. The RFP also requires the solution to be **cloud-ready, private/public cloud deployable**, and capable of running in OICL's environment while meeting regulatory and security requirements.

1. Overall Cloud Segregation

USERS

Business Users

Executives

Fraud Analysts

Data Scientists

Branch Offices

Regional Offices

External Agencies

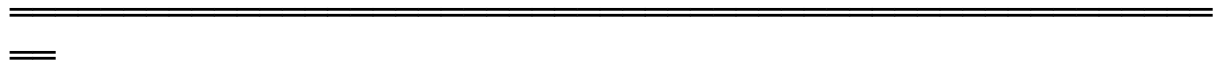
APIs



GOVERNMENT COMMUNITY CLOUD (GCC)

- 1. Consumption Layer
- 2. AI & Analytics Layer
- 3. Lakehouse

4. Governance Layer
5. Data Ingestion
6. Security
7. Infrastructure
8. Monitoring
9. Backup
10. Disaster Recovery



OICL Internal Systems

External Agencies

Hospitals

TPA

Police

IIB

SAP

CRM

INLIAS

Everything except the users resides inside the secured GCC environment.

2. What should be inside the Consumption Layer?

This is the **only layer that business users interact with.**

A. Executive Dashboards

Used by

- CMD
- Directors
- CIO
- Business Heads

Examples

- Premium Collection
 - Claims
 - Fraud Loss
 - Customer Growth
 - KPI Dashboard
 - Operational Dashboard
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B. Business Intelligence (BI)

Tools

- Power BI
- Tableau
- Qlik
- Superset

Capabilities

- Reports
 - Drill-down
 - Slice & Dice
 - Trend Analysis
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C. Self-Service Analytics

Users

- Business Analysts

- Operations Team

Capabilities

- Create reports
 - Run SQL
 - Build charts
 - Ad hoc analysis
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D. Fraud Investigation Portal

Shows

- Fraud Alerts
 - Risk Scores
 - Explainable AI
 - Case Management
 - Investigation History
-

E. AI Applications

Examples

- Claim prediction
 - Fraud prediction
 - Customer segmentation
 - Cross-selling
 - Churn prediction
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F. Data APIs

Consumed by

- Mobile Apps
- Web Portal
- CRM

- Core Insurance
- Third-party systems

Examples

Customer API

Policy API

Claims API

Fraud API

Risk Score API

Premium API

G. Reports

Operational Reports

MIS Reports

IRDAI Reports

Compliance Reports

Audit Reports

Monthly Reports

Quarterly Reports

Annual Reports

H. Alerts & Notifications

Examples

High-risk fraud

Claim anomaly

Data quality issue

Pipeline failure

Threshold breach

Business KPI alerts

I. External Portal

If required

Government agencies

TPA

Hospitals

Partner access

(Read-only through secure APIs.)

Final Consumption Layer

Consumption Layer

Executive Dashboards

Business Intelligence

Self-Service Analytics

Reports

Fraud Portal

AI Applications

Data APIs

Alerts

Notifications

Partner Portal

Mobile Access

Role Based Access

3. What should be inside GCC?

Everything below is hosted **inside the Government Community Cloud**.

Infrastructure

Virtual Machines

Kubernetes

Docker

Object Storage

File Storage

Load Balancer

Network

Firewall

VPN

Compute

Spark Cluster

SQL Engine

Streaming Engine

ML Compute

GPU

Containers

Storage

Bronze

Silver

Gold

Object Storage

Metadata Repository

Feature Store

Backup Storage

Archive Storage

Governance

Metadata

Catalog

Lineage

MDM

Data Quality

Glossary

Policy Engine

Audit Logs

AI Platform

ML Workbench

Notebook

Model Registry

Feature Store

Training

Inference

Explainable AI

Data Processing

Spark

Kafka

CDC

Streaming

Batch

Transformation

Aggregation

Security

IAM

RBAC

MFA

Encryption

Data Masking

DLP

SIEM

SOAR

HSM

KMS

PKI

Secrets Management

Monitoring

Application Monitoring

Infrastructure Monitoring

Cluster Monitoring

Log Analytics

Alert Manager

Performance Dashboard

Backup

Snapshots

Replication

Object Backup

Database Backup

Recovery Vault

Disaster Recovery

Secondary GCC Region

Replication

Failover

Backup

Recovery Automation

4. Network Segregation inside GCC

INTERNET



WAF / API Gateway



Load Balancer



Presentation Network



Dashboards

Reports

APIs

Web Portal



Analytics Network



Spark

Kafka

ML

Notebook

Feature Store

Data Network

Bronze

Silver

Gold

Metadata

MDM

Object Storage

Management Network

Monitoring

Logging

Backup

Kubernetes

CI/CD

5. End-to-End Data Flow

OICL Systems

|



Data Ingestion

(Batch / Streaming / APIs)

|



Bronze Zone

(Raw Data)

|



Silver Zone

(Cleansed Data)

|



Gold Zone

(Business Data)

|



Analytics
AI Models
Fraud Detection
|
▼
Dashboards
Reports
APIs
Alerts
Business Users

Recommended Segregation for an OEM Architecture

Layer	Hosted in GCC?	Accessible by Users?
Source Systems	Partly (OICL DC/External)	No
Data Ingestion	✓ Yes	No
Data Lakehouse (Bronze/Silver/Gold)	✓ Yes	No
Governance & Metadata	✓ Yes	No
AI/ML Platform	✓ Yes	Limited (Data Scientists)
Processing & Analytics	✓ Yes	No
Security Services	✓ Yes	Admins only
Monitoring & Operations	✓ Yes	Admins only
Backup & DR	✓ Yes	No
Consumption Layer (Dashboards, Reports, APIs, Alerts)	✓ Yes	Yes

This segregation aligns with the RFP's objectives of a centralized, secure Data Lakehouse with governance, analytics, fraud detection, AI/ML, APIs, and cloud readiness, while ensuring that only the presentation and consumption services are directly exposed to end users.

For this RFP, the **Consumption Layer** is the topmost layer of the architecture where the processed, governed, and secured data is finally consumed by business users, applications, and external systems. The project objective also mentions that the platform should expose analytics services, master data services, dashboards, reports, AI outputs, and APIs for business consumption.

Consumption Layer

The Consumption Layer should include the following components:

Business Users

- Business Executives
- Regional Offices
- Branch Offices
- Claims Officers
- Underwriters
- Finance Team
- Operations Team

Analytics Consumers

- Data Analysts
- Business Intelligence Teams
- Fraud Investigation Teams
- Risk Management Teams
- Actuarial Teams
- Compliance Teams

Data Science Consumers

- Data Scientists
- Machine Learning Engineers
- AI Researchers
- Citizen Data Scientists

Business Applications

- Interactive Dashboards

- Operational Reports
- Self-Service BI
- KPI Dashboards
- Executive Cockpits
- Mobile Dashboards

AI & Advanced Analytics

- Fraud Detection Portal
- Risk Scoring Engine
- Customer Segmentation
- Predictive Analytics
- Explainable AI Results
- Recommendation Engine

Enterprise Integration

- REST APIs
- GraphQL APIs (if supported)
- Data Services
- Common Data Exchange Hub
- Event APIs
- External Partner APIs

Notifications

- Email Alerts
- SMS Alerts
- Push Notifications
- Workflow Notifications

Cloud Segregation (Government Community Cloud)

The RFP requires that the solution be **cloud-ready**, support deployment on **private and public cloud**, and use a **Government Community Cloud (GCC)** model where

applicable. The architecture should therefore logically separate workloads based on security, compliance, and operational requirements.

1. OICL On-Premise / Private Cloud

The most sensitive business systems remain within OICL's own infrastructure. This includes the primary and disaster recovery data centres, the core insurance application (INLIAS), SAP, HRMS, claims processing, identity services, and other systems of record. Raw operational data is generated here before being securely transferred into the Lakehouse environment. Keeping these workloads in the private cloud ensures direct control over critical business operations, regulatory compliance, and integration with existing infrastructure.

2. Government Community Cloud (GCC)

The Government Community Cloud hosts the enterprise Data Lakehouse and advanced analytics platform while meeting government security and data residency requirements. This is where most data engineering, governance, analytics, and AI services execute.

Typical services in the GCC include:

- Data Ingestion Services
- Enterprise Data Lakehouse
- Bronze, Silver and Gold storage zones
- Data Processing Engines (Spark/Flink)
- Streaming Services (Kafka or equivalent)
- Metadata Repository
- Data Catalog
- Data Quality Engine
- Data Lineage
- Master Data Management (MDM)
- AI/ML Workbench
- Feature Store
- Model Training and Deployment
- Business Intelligence Platform
- Dashboard and Reporting Services

- API Gateway
- Common Data Exchange Hub
- Monitoring and Logging
- Backup Services
- Disaster Recovery replication
- Security services such as IAM, RBAC, encryption, key management, SIEM, SOAR, DLP, and audit logging

The GCC becomes the central analytics platform where enterprise data is consolidated, governed, processed, and securely exposed to authorised users.

3. External Systems Zone

The external integration zone connects trusted third-party organisations without exposing internal systems directly. This includes:

- Insurance Information Bureau (IIB)
- Third Party Administrators (TPAs)
- Hospitals
- Police Departments
- Government databases
- Regulatory agencies
- Banking and payment gateways
- External fraud intelligence providers
- Other approved partners

These systems exchange data through secure APIs, managed file transfers, or streaming interfaces via the API Gateway and Common Data Exchange Hub.

4. User Access Zone

Business users do not connect directly to the Data Lakehouse. Instead, they access dashboards, reports, self-service analytics, AI outputs, and APIs through secure authentication mechanisms such as IAM, RBAC, and MFA. This logical separation

protects the underlying data platform while allowing controlled access based on user roles.

End-to-End Cloud Segregation Flow

OICL Private Cloud / Data Centres

Core Insurance (INLIAS)

Claims System

SAP

HRMS

CRM

ECM

Email

Policy Administration

Identity Services

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Secure Connectivity (VPN/MPLS/API)

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Government Community Cloud (GCC)

Data Ingestion

Streaming

Enterprise Data Lakehouse

Data Processing

Governance

Metadata

Data Catalog

Data Lineage

MDM

AI/ML Platform

BI Platform

Dashboards

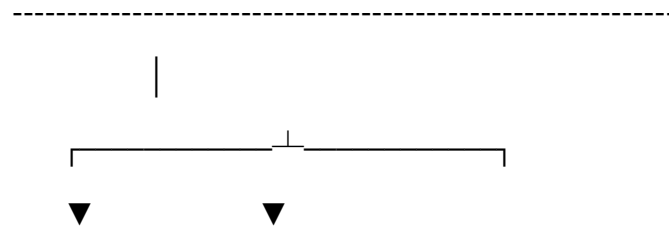
API Gateway

Security Services

Monitoring

Backup

Disaster Recovery



Business Users

External Partners

Dashboards

IIB

Reports

TPAs

APIs

Hospitals

AI Results

Police

Self-Service BI

Regulators

This segregation provides a clear security boundary: **transactional systems remain in OICL's controlled environment, the GCC hosts the scalable analytics and governance platform, and all user and partner access is mediated through secure APIs and role-based controls.**



Core Insurance (INLIAS)
 Claims System
 SAP
 HRMS
 CRM
 ECM
 Email
 Policy Administration
 Identity Services

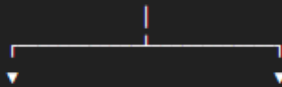


Secure Connectivity (VPN/MPLS/API)



Government Community Cloud (GCC)

Data Ingestion
 Streaming
 Enterprise Data Lakehouse
 Data Processing
 Governance
 Metadata
 Data Catalog
 Data Lineage
 MDM
 AI/ML Platform
 BI Platform
 Dashboards
 API Gateway
 Security Services
 Monitoring
 Backup
 Disaster Recovery



Business Users
 Dashboards
 Reports
 APIs
 AI Results
 Self-Service BI

External Partners
 IIB
 TPAs
 Hospitals
 Police
 Regulators

